

PHITRON AI/ML CAREER TRACK

AI & ML ENGINEER ROADMAP

A complete step-by-step guide to become an AI & ML Engineer from scratch.
Only free resources used!

ABOUT THIS ROADMAP

This roadmap is built for absolute beginners who want to become AI & ML Engineers from scratch. Every single resource listed here is **100% free**—you do not need to buy any paid courses or certifications.

The roadmap is divided into **7 easy-to-follow phases** across **12 months** of consistent daily study. Make sure to follow the phases in order and do not skip any phase. The basics you learn in Phase 1 are very important to understand the advanced topics in Phase 3 and beyond.

By the end of Month 12, you will have a strong understanding of how ML works, build and deploy 3 to 5 real-world projects, and get completely ready to clear job interviews at startups and top tech companies.

12-MONTH TIMELINE OVERVIEW

Timeline	Focus Areas & Core Curricula	Phase Track
Month 1-2	Foundational Mathematics, Core Python Programming, NumPy, and Pandas	Phase 1

Timeline	Focus Areas & Core Curricula	Phase Track
Month 3-4	Classical Machine Learning Algorithms, Scikit-Learn Framework, and Initial Kaggle Formations	Phase 2
Month 5-7	Deep Learning Foundations, PyTorch Architecture, Advanced CNNs, and Transformers	Phase 3
Month 8-9	Domain Specialization Track (Natural Language Processing / Computer Vision / Production MLOps)	Phase 4
Month 10	Production MLOps Engineering, Docker Containerization, FastAPI Architectures, and System Deployment	Phase 5
Month 11	Public Portfolio Deployment, Competitive Kaggle Engineering, and Technical Writing Output	Phase 6
Month 12	Advanced Technical Interview Drills, DSA LeetCode Focus, Applications, and Strategic Mock Interviewing	Phase 7

Phase 1: Foundations (Math & Python)

Timeline: Months 1-2

Goal: Learn the essential math and programming basics that every ML engineer needs. Do not skip this part—everything else in ML is built on these core skills.

Core Topics to Master

- **Linear Algebra:** Vectors, matrices, vector space transformations, matrix multiplication, eigenvalues/eigenvectors, dot products, and Singular Value Decomposition (SVD).
- **Calculus:** Function limits, scalar derivatives, partial derivatives, multivariate chain rules, gradients, and optimization theory.
- **Statistics & Probability:** Descriptive metrics (mean, median, variance), distinct probability distributions, Bayes' Theorem, and rigorous hypothesis testing.
- **Python Ecosystem:** Comprehensive execution of NumPy for numerical structures, Pandas for structural data analysis, and Matplotlib for analytical visualization.
- **Software Fundamentals:** Modern procedural execution, object-oriented programming (OOP), file input/output management, pip dependencies, and isolated virtual environments.

Acquired Competencies

PYTHON

NUMPY

PANDAS

MATPLOTLIB

LINEAR ALGEBRA

PROBABILITY & STATS

Free Learning Resources

3Blue1Brown - Essence of Linear Algebra

youtube.com/c/3blue1brown

Elite geometric and visual intuition for linear equations and multi-dimensional matrices.

Khan Academy - Statistics & Probability

khanacademy.org/math/statistics-probability

Beginner-to-intermediate exploration covering essential statistical foundations.

Python for Everybody (Dr. Chuck - Coursera Audit)

coursera.org/specializations/python

A gold-standard introduction to program design, loops, data architecture, and processing.

Harvard CS50P - Introduction to Programming with Python

cs50.harvard.edu/python

Excellent structured course mapping out production-level programming paradigms.

Kaggle Micro-Courses & Documentation Docs

kaggle.com/learn

Interactive, cloud-hosted notebooks optimized for fast adoption of Pandas, NumPy, and charting.

StatQuest with Josh Starmer

youtube.com/c/joshstarmer

Highly accessible, highly structured step-by-step breakdowns of complex statistical models.

Hands-On Milestone Projects

P1: Construct a thorough data-manipulation script implementing Pandas to parse, clean, and filter real-world raw CSV datasets from Kaggle.

P2: Generate comprehensive exploratory data visualizations (including bar plots, distributions, and multi-axis scatter plots) leveraging Matplotlib.

P3: Resolve 20 engineering challenges sequentially via the HackerRank Python track to reinforce structural syntax and operational logic.

Phase 2: Machine Learning Fundamentals

Timeline: Months 3-4

Goal: Learn the core machine learning algorithms. Focus completely on understanding how and why each algorithm works, instead of just copying library code.

Core Topics to Master

- **Supervised Learning:** Linear regression, logistic optimization, decision tree mechanics, Random Forests, Support Vector Machines (SVM), and K-Nearest Neighbors (KNN).
- **Unsupervised Learning:** Coordinate clustering (K-Means), density-based clustering (DBSCAN), and Principal Component Analysis (PCA) for structural dimensionality reduction.
- **Model Evaluation Architecture:** Stratified train/test partitioning, k-fold cross-validation, confusion matrices, precision-recall optimization, F1-scores, and ROC-AUC properties.
- **Overfitting Mitigation:** The mathematical bias-variance tradeoff curve, **L1** and **L2** weight regularization frameworks, and early execution limits.
- **Advanced Feature Engineering:** Categorical encoding pipelines, Z-score normalization, min-max feature scaling, advanced missing-value imputation, and statistical feature selection.
- **Scikit-Learn Infrastructure:** Master transformation pipelines, preprocessing abstraction layers, and hyperparameter grids.

Acquired Competencies

SCIKIT-LEARN

REGRESSION

CLASSIFICATION

CLUSTERING

CROSS-VALIDATION

FEATURE ENGINEERING

Free Learning Resources

Andrew Ng - Machine Learning Specialization (Coursera Audit)

coursera.org/specializations/machine-learning-introduction

The globally recognized classic covering fundamental machine learning and practical application mechanics.

Google Machine Learning Crash Course

developers.google.com/machine-learning/crash-course

Highly practical, fast-paced technical guide pairing concepts with direct code implementations.

Scikit-Learn Official Tutorials & Documentation

scikit-learn.org/stable/tutorial

The absolute standard for structural pipeline engineering and data transformation execution.

Kaggle Learn ML Pathways

kaggle.com/learn/intro-to-machine-learning

Hands-on micro-courses built with immediate cloud execution environments to cement theoretical concepts.

Made With ML by Goku Mohandas

madewithml.com

Excellent end-to-end framework merging pure ML theory with concrete, engineering-first implementations.

Hands-On Milestone Projects

P1: Program and hyper-tune a real estate valuation engine utilizing linear and multivariate regression workflows.

P2: Build an automated text classification matrix (e.g., Email Spam Classifier) using logistic regression or Naive Bayes architectures.

P3: Execute unsupervised consumer clustering by mapping dimensional profiles via K-Means or DBSCAN parameters.

P4: Launch, structure, and submit your initial predictive model to the legendary introductory Kaggle competition (Titanic Survival Matrix).

Phase 3: Deep Learning & Neural Networks

Timeline: Months 5-7

***Goal:** Master deep learning—the technology behind modern AI. Learn how to easily build, train, and fix neural networks using PyTorch and TensorFlow.*

Core Topics to Master

- **Neural Architectures:** Dense multi-layer perceptrons, mathematical activation functions (ReLU, GELU, Softmax), forward evaluation, backpropagation, and stochastic gradient descent optimization.
- **Convolutional Neural Networks (CNNs):** Spatial kernel operations, pooling strategies, structural image classification pipelines, and deep transfer learning paradigms using pre-trained weights (ResNet, VGG).
- **Sequence Models (RNNs & LSTMs):** Temporal sequence alignment, time-series operations, and handling vanishing/exploding gradient states.
- **Attention Mechanisms & Transformers:** Self-attention formulations, scaled multi-head vector mapping, tokenization processes, and the primary building blocks of structural GPT/BERT systems.
- **PyTorch Ecosystem:** Dynamic graph building, multi-dimensional tensor matrix tracking, automatic differentiation (autograd), custom Dataset/DataLoader pipelines, and granular raw training loops.
- **TensorFlow & Keras:** Declarative Sequential and Functional graph APIs, structural custom training hooks, and modular optimization configurations.
- **Optimization Mechanics:** Execution of Adam, AdamW, and RMSprop optimizers; batch normalization layers; dropout distribution; and multi-step learning-rate schedulers.

Acquired Competencies

PYTORCH

TENSORFLOW

KERAS

CNNS

RNNS

TRANSFORMERS

Free Learning Resources

fast.ai - Practical Deep Learning for Coders

fast.ai

Renowned top-down pedagogical style focusing heavily on rapid state-of-the-art results and intuition.

Deep Learning Specialization (Andrew Ng - Coursera Audit)

coursera.org/specializations/deep-learning

The absolute global industry benchmark covering structural vector calculus, hyperparameter tuning, and model optimization.

PyTorch Official Guides & Video Tutorials

pytorch.org/tutorials

The most direct approach for mastering foundational object architectures and training configurations.

Andrej Karpathy - Neural Networks: Zero to Hero Series

youtube.com/@AndrejKarpathy

Unrivaled masterclass breaking down raw backpropagation, building custom GPT models from scratch, and structural micrograd logic.

Stanford CS231n - Convolutional Neural Networks for Visual Recognition

cs231n.stanford.edu

In-depth technical lecture notes, math matrices, and computing logic for high-performance vision models.

The Illustrated Transformer (Jay Alammar)

jalammar.github.io/illustrated-transformer

Elite graphical and conceptual breakdown detailing the mechanics of scaled self-attention vectors.

Hands-On Milestone Projects

P1: Construct a binary or multi-class visual computer vision system (e.g., Cat vs. Dog recognition matrix) using PyTorch convolutional configurations.

P2: Design and train an LSTM sequential system to execute text sentiment parsing across large-scale movie evaluation metrics.

P3: Implement advanced visual feature extraction via fine-tuning pre-trained weights (ResNet50) onto niche, highly specialized custom structural image classes.

P4: Design a generative or autoregressive sequential system to stream predictive text block synthesis character-by-character.

Phase 4: Domain Specialization Track

Timeline: Months 8-9

Goal: Choose EXACTLY ONE specialization track to focus on. Companies prefer hiring developers with solid skills in one area for junior positions instead of someone who knows a little bit of everything.

Available Technical Vectors (Choose One)

Track A: Natural Language Processing & Large Language Models (NLP & LLMs)

Advanced string tokenization, structural vector embedding mapping, custom encoder-decoder fine-tuning (BERT, T5), Retrieval-Augmented Generation (RAG) framework construction, orchestration engines (LangChain/LlamaIndex), and modern contextual prompt execution engineering.

Track B: Computer Vision (CV)

Advanced object localization and real-time inference (YOLO series), pixel-level semantic image segmentation, digital image matrix processing via OpenCV, Generative Adversarial Networks (GANs), and latent diffusion mapping concepts.

Track C: Machine Learning Operations (MLOps & Production Engineering)

High-availability model serving strategies, immutable container orchestration (Docker), structured high-throughput REST APIs (FastAPI), continuous logging and operational model health metrics, unified CI/CD integration, and active feature storage systems.

Track D: Advanced Data Science & Analytical Intelligence

Advanced multi-factor exploratory data analysis (EDA), rigorous digital A/B testing frameworks, structural multivariate autoregressive time-series projection (ARIMA, Prophet), and scalable business enterprise analytics architecture.

Ecosystem & Tool Abstractions

HUGGINGFACE HUB

LANGCHAIN

OPENCV

FASTAPI

DOCKER CONTAINERS

YOLO ARCHITECTURES

MLFLOW TRACKING

Free Learning Resources

Hugging Face NLP Core Pathway

huggingface.co/learn/nlp-course

The primary course for working with large language models, datasets, and tokenization tools.

LangChain Architecture Documents & Developer Bootcamps

python.langchain.com/docs

Comprehensive structural guide detailing how to link vector data indexes with Large Language Models.

OpenCV Python Development Core Tutorials

docs.opencv.org/4.x/d6/d00/tutorial_py_root.html

Essential reference detailing advanced pixel matrix operations, filtering techniques, and frame handling.

Full Stack Deep Learning Production Course

fullstackdeeplearning.com

Premier curriculum tracking how to step smoothly from single local notebooks to deployed production environments.

Roboflow Visual Ecosystem Blog & Guides

blog.roboflow.com

Excellent technical guide detailing custom annotation workflows, data augmentation, and modern object detection.

MLflow & DVC Engineering Documentations

mlflow.org / dvc.org

The industry standard for model checkpoint indexing, parameters log tracking, and strict data versioning.

Hands-On Domain Milestone Projects

P1 (NLP Focus): Engineer an isolated Retrieval-Augmented Generation (RAG) conversational interface fine-tuned on custom PDFs via Hugging Face.

P2 (Vision Focus): Program and fine-tune a high-speed real-time localized camera system applying custom trained YOLOv8 nodes.

P3 (MLOps Focus): Wrap an analytical deep learning model inside an immutable Docker image layer, serving API request loops with FastAPI.

P4 (Data Science Focus): Formulate a highly parameter-optimized time-series forecasting model using Prophet to predict multi-variable seasonal vectors.

Phase 5: MLOps, Tools & Production Systems

Timeline: Month 10

Goal: Learn how real ML systems run inside engineering teams. Knowing how to deploy, monitor, and scale models is what makes you fully job-ready.

Core Topics to Master

- **Version Control & CI/CD:** Modular git structures, collaborative pull request architectures, and declarative continuous integration automated pipelines via GitHub Actions.
- **Container Infrastructure:** Writing efficient multi-stage Dockerfiles, constructing multi-container topologies using docker-compose, and optimizing visual image weight.
- **High-Throughput Web Protocols:** Creating asynchronous Python web services via FastAPI or Flask, handling custom payload validation, and structuring error handling layers.
- **Experimentation Registry:** Running analytical experiment logging matrices with MLflow or Weights & Biases (W&B) to catalog precision tracking, artifact variables, and weights.
- **Cloud Provisioning:** Provisioning isolated computational virtual machines, mapping out storage bucket systems, and scaling auto-managed containers via Google Cloud Run or AWS instances.
- **Relational SQL Query Engines:** Designing multi-layer relational data extractions, writing nested analytical window equations, and configuring optimized analytical queries for upstream processing pipelines.

Acquired Competencies

GIT & GITHUB

DOCKER CONTAINERS

FASTAPI SERVICES

MLFLOW LOGGING

RELATIONAL SQL

GCP / AWS RUNNING

CI/CD TRACKING

Free Learning Resources

Git & GitHub Comprehensive Course (freeCodeCamp)

youtube.com/c/freecodecamp

Deep immersion into command-line version tracking, remote branch management, and continuous automation.

Docker for Absolute Beginners (TechWorld with Nana)

youtube.com/@TechWorldwithNana

The gold-standard video introduction for mastering micro-container architectures, networking, and imaging.

FastAPI Production Documentation System

fastapi.tiangolo.com/tutorial

Highly structured documentation demonstrating input verification parsing, async processes, and OpenAPI configuration.

MLflow Fast Experimentation Quickstart

mlflow.org/docs/latest/quickstart.html

Accelerated workflow guide showing how to quickly spin up localized parameters and weights logging dashboards.

Weights & Biases (W&B) Technical Academy

wandb.ai/courses

Elite learning sequences covering visual metric instrumentation and complex hyperparameter tracking.

Mode Analytics Interactive SQL Tracks

mode.com/sql-tutorial

Hands-on SQL code environments covering complex table filters, joins, and database queries.

Hands-On Production Milestone Projects

P1: Enclose a weights-optimized deep learning inference pipeline inside an immutable Docker image, serving API endpoints using FastAPI validation layers.

P2: Integrate an automated local MLflow logging dashboard contextually into an arbitrary machine learning processing script.

P3: Package an application engine into a serverless cloud instance (e.g., Google Cloud Run), tracking and stress-testing the live public URI endpoint.

P4: Design an automated data system: execute data pulling via SQL -> clean features -> auto-train weights -> track via registry -> live deploy.

Phase 6: High-Value Public Portfolio Architecture

Timeline: Month 11

Goal: Build 3 to 5 strong portfolio projects that show your real skills. Most companies will evaluate your skills based on your projects—your GitHub profile is your true resume!

Core Core Curricula Focus Areas

- **Complete End-to-End Production Systems:** Build systems tracking the entire pipeline from automated raw data fetching to operational system serving.
- **GitHub Repository Polish:** Write explicit, informative Markdown README files featuring architectural flow diagrams, system instructions, and live performance metrics.
- **Competitive Kaggle Engineering:** Enter ongoing or historical intermediate Kaggle tracks with the specific goal of achieving a Top 20% ranking.
- **Academic Deep-Dive Implementation:** Read formal AI research papers (arXiv) and implement the underlying architectures from scratch in clean, modular Python.
- **Streamlit Interface Prototyping:** Wrap complex backend deep learning frameworks within elegant, interactive web dashboards for stakeholders.
- **Technical Writing & Communication:** Write clear, deep-dive technical breakdowns mapping out engineering details for personal or corporate blogs.

Acquired Competencies

STREAMLIT LAYOUT

GITHUB ENGINEERING

KAGGLE POSITIONING

PAPER EXECUTION

SYSTEM DOCUMENTATION

TECHNICAL BLOGGING

Free Learning Resources

Streamlit Application Core Documentations

docs.streamlit.io

Excellent operational reference guide for transforming raw Python code into production-grade user dashboards in minutes.

Papers With Code Database Portal

paperswithcode.com

The largest curated collection connecting indexed scientific papers with their explicit open-source repository frameworks.

Kaggle Active Competition Platforms

kaggle.com/competitions

Real-world operational tracks to challenge pipeline efficiency and feature engineering under constraints.

GitHub Developer Education Pack

education.github.com/pack

High-value ecosystem benefits for verified students, unlocking free cloud infrastructure credits and tooling.

Arxiv Repository (Machine Learning Track)

arxiv.org/list/cs.LG/recent

The open global source for daily research paper releases tracking the absolute bleeding edge of artificial intelligence.

Hands-On Portfolio Milestone Projects

P1 (NLP): Deploy a conversational document query interface (RAG) running live via Streamlit Cloud.

P2 (Vision): Build a high-throughput video object detection system that seamlessly processes webcam frame inputs.

P3 (End-to-End): Architect an end-to-end pipeline that ingests data automatically, runs continuous feature processing, updates models, and hosts a live endpoint.

P4 (Kaggle): Secure a verified top-20% ranking in any active intermediate Kaggle data track.

P5 (Blogging): Publish 3 technical deep-dives on Medium or Dev.to explaining your project architectures.

Phase 7: Interview Preparation & Job Placement Strategy

Timeline: Month 12

Goal: Crack your AI & ML job interviews! Companies will test your ML theory, coding skills (DSA), and hands-on project knowledge. Practice all three areas together to perform your best.

Core Topics to Master

- **Theoretical Machine Learning Inquiries:** Articulate the absolute inner mechanics of structural bias/variance optimization, loss equations, gradient descent convergence, cross-entropy foundations, and regularized parameter limits.
- **Data Structures & Algorithms (DSA):** Master arrays, nested string indices, optimized HashMaps, structural trees, sorting, BFS/DFS traversal loops, and fundamental dynamic programming via LeetCode Easy/Medium tiers.
- **Production SQL Verification:** Writing complex database queries, handling nested subqueries, and grouping metrics under strict execution plans.
- **Enterprise AI System Design:** Formulating high-availability architectures for real-time recommendations, enterprise search ranking structures, and fraud detection alert loops.
- **Behavioral Communication (STAR Protocol):** Mapping out personal engineering milestones using the explicit Situation, Task, Action, and Result matrix. Prepare 5 distinct, high-impact project stories.
- **Resume & Public Profile Tailoring:** Restructuring personal portfolios down to a single high-impact page, clearly showcasing live project links, deployment metrics, and clean code repositories.
- **Simulated Interview Execution:** Conducting live, peer-to-peer technical interview sessions under realistic constraints.

Acquired Competencies

LEETCODE DSA

ML ARCHITECTURE THEORY

PRODUCTION SQL

AI SYSTEM DESIGN

STAR BEHAVIORAL

RESUME LAYOUT

Free Learning Resources

LeetCode Algorithmic Frameworks (Free Tier)

leetcode.com

The absolute standard for practicing technical data structures, arrays, searches, and loops.

Ace the ML Interview Repositorygithub.com/khangich/machine-learning-interview

An incredible curation of over 500+ verified machine learning engineering questions with comprehensive answers.

Machine Learning Cheat Sheet Guideml-cheatsheet.readthedocs.io

Excellent online reference guide summarizing core optimization mathematics, gradients, and model architectures.

Pramp Free Peer-to-Peer Mock Interviewspramp.com

Excellent automated platform to practice real-time technical coding and system design with peer engineers.

LeetCode Database Core Problem Setleetcode.com/problemset/database

Highly targeted SQL problem track focusing on joins, aggregations, and query optimizations.

Harvard CS50 System Design Compendiumscs.harvard.edu/~ctq

Elite foundational analysis detailing horizontal/vertical scaling, database optimization, caching, and infrastructure.

interviewing.io (Community Sessions)interviewing.io

Excellent platform offering free anonymous mock interviews with seasoned industry engineers.

Hands-On Milestone Projects

P1: Solve 100 targeted LeetCode algorithm problems (70 Easy, 30 Medium) covering core patterns.

P2: Complete 10 live simulated technical interviews on Pramp, recording and reviewing your communication structure.

P3: Track applications to 5 targeted AI/ML engineering roles daily using a detailed spreadsheet pipeline.

P4: Connect and engage with 3 working ML engineers weekly on LinkedIn to request technical reviews and potential referrals.

8 PRO TIPS FOR YOUR AI/ML JOURNEY

1. Consistency beats intensity

Study for 2 to 3 hours every day instead of studying for 10 hours straight on weekends. ML is like a marathon; daily practice helps you build solid skills that last.

2. Build projects from Day 1

Do not wait until you feel completely ready to start making projects. Build small tools right from Phase 1. Employers care about what you have built, not just what you studied.

3. Use Google Colab for free GPUs

Google Colab gives you free access to powerful GPUs to train your deep learning models. You do not need to buy an expensive computer. Use it for your neural network models from Phase 3 onward.

4. Kaggle is your best practice ground

Kaggle has tons of free datasets, interactive notebooks, and free cloud compute. Join competitions early. Even a top 50% finish is a fantastic addition to your beginner resume.

5. Network actively on LinkedIn

Follow top ML developers, share updates about your ongoing projects, and leave helpful comments on AI posts. Around 70% of tech roles are filled through direct referrals and LinkedIn networking.

6. Read 1 ML paper per month from Phase 3

Go to paperswithcode.com and find a research paper that interests you. Read the Abstract, Introduction, and Conclusion first, then try to write or run the code. This will build deep expertise quickly.

7. Focus on ONE specialization track

When you reach Phase 4, pick exactly one track and master it. Being excellent at either NLP OR Computer Vision makes you much more hireable than being average at both. Depth beats breadth for fresh graduates.

8. Keep your GitHub profile clean and active

Engineering managers check your repositories before setting up an interview. Write clear README files with project screenshots, add simple demo GIFs, organize your project files, and commit code regularly.

JOB INTERVIEW PREPARATION CHECKLIST

Use this checklist during Month 12. Try to check off every single box before sending your very first job application.

Technical Core Preparation

- ☐ Resolve 100 comprehensive LeetCode data problems (Arrays, Matrices, HashMaps, Trees, Dynamic Programming).
- ☐ Internalize and articulate 50 foundational machine learning theoretical questions and mathematical optimizations.
- ☐ Practice pitching your project architectures out loud, summarizing the technical details clearly in under 3 minutes.
- ☐ Resolve 30 database query engineering questions from the LeetCode repository.
- ☐ Study 3 major enterprise AI system design challenges (Recommendation Matrix, Scale Search Ranking, Real-time Fraud Detection).

Portfolio & Presentation Checklist

- ☐ Ensure your GitHub repository features at least 3 production-grade, highly documented project folders with clean code.
- ☐ Verify at least 1 complex project is completely hosted and accessible live via a public link.
- ☐ Tailor your LinkedIn profile to highlight your technical projects, accompanied by links and structural summaries.
- ☐ Format a polished, single-page professional resume tailored specifically to AI/ML engineering roles, featuring your GitHub link.
- ☐ Memorize a refined, 2-minute elevator pitch detailing the core architecture, constraints, and solutions for each project.

Market Application Strategy

- ☐ Submit your tailored profile to 5 matching AI/ML roles daily—targeting high-growth startups is often an effective strategy for entering the field.

- [] Filter job pipelines specifically for: Machine Learning Engineer, Junior AI Engineer, Data Scientist, or Applied Research Engineer.
- [] Combine LinkedIn Easy Apply submissions with direct, personalized cold emails to engineering managers containing your portfolio links.
- [] Expand your search globally; remote AI/ML engineering positions offer vast opportunities for distributed engineering teams worldwide.
- [] Follow up every application within 24 hours by reaching out to the respective hiring managers on LinkedIn.

The AI and Machine Learning space rewards hard work above everything else. Everyone finds the math tough at the beginning—that is completely normal!

Follow this roadmap step by step, build every project, and you WILL be job-ready in 12 months. All these learning resources are completely free. The only cost is your time, focus, and consistency.